

Verify that Telnet, HTTP, and HTTPS are disabled and that the SSH port is open:

```
[(base) timjan@Mac ~ % nc -vz 192.168.99.11 80
nc: connectx to 192.168.99.11 port 80 (tcp) failed: Connection refused
[(base) timjan@Mac ~ % nc -vz 192.168.99.11 443
nc: connectx to 192.168.99.11 port 443 (tcp) failed: Connection refused
[(base) timjan@Mac ~ % nc -vz 192.168.99.12 80
nc: connectx to 192.168.99.12 port 80 (tcp) failed: Connection refused
[(base) timjan@Mac ~ % nc -vz 192.168.99.12 443
nc: connectx to 192.168.99.12 port 443 (tcp) failed: Connection refused
[(base) timjan@Mac ~ % nc -vz 192.168.99.11 22
Connection to 192.168.99.11 port 22 [tcp/ssh] succeeded!
[(base) timjan@Mac ~ % nc -vz 192.168.99.12 22
Connection to 192.168.99.12 port 22 [tcp/ssh] succeeded!

[(base) timjan@Mac ~ % nc -vz 192.168.99.11 23
nc: connectx to 192.168.99.11 port 23 (tcp) failed: Connection refused
[(base) timjan@Mac ~ % nc -vz 192.168.99.11 22
Connection to 192.168.99.11 port 22 [tcp/ssh] succeeded!
[(base) timjan@Mac ~ % nc -vz 192.168.99.12 23
nc: connectx to 192.168.99.12 port 23 (tcp) failed: Connection refused
[(base) timjan@Mac ~ % nc -vz 192.168.99.12 22
Connection to 192.168.99.12 port 22 [tcp/ssh] succeeded!
(base) timjan@Mac ~ %
```

Move the MacBook-facing switch port from VLAN 99 to VLAN 10:

```
[(base) timjan@Mac ~ % ping 192.168.99.11
PING 192.168.99.11 (192.168.99.11): 56 data bytes
Request timeout for icmp_seq 0
Request timeout for icmp_seq 1
Request timeout for icmp_seq 2
Request timeout for icmp_seq 3
ping: sendto: No route to host
Request timeout for icmp_seq 4
^C
--- 192.168.99.11 ping statistics ---
6 packets transmitted, 0 packets received, 100.0% packet loss
```

Move the MacBook-facing switch port back from VLAN 10 to VLAN 99:

```
[(base) timjan@Mac ~ % ping 192.168.99.11
PING 192.168.99.11 (192.168.99.11): 56 data bytes
64 bytes from 192.168.99.11: icmp_seq=0 ttl=255 time=3.759 ms
64 bytes from 192.168.99.11: icmp_seq=1 ttl=255 time=2.950 ms
64 bytes from 192.168.99.11: icmp_seq=2 ttl=255 time=2.895 ms
64 bytes from 192.168.99.11: icmp_seq=3 ttl=255 time=3.093 ms
64 bytes from 192.168.99.11: icmp_seq=4 ttl=255 time=2.659 ms
^C
--- 192.168.99.11 ping statistics ---
5 packets transmitted, 5 packets received, 0.0% packet loss
```

SW1:

```
SW1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/1, Gi0/2
10	TEST_CLIENTS	active	
99	MGMT	active	Fa0/2
999	PARKING_LOT	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24

```
SW1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	99

Port	Vlans allowed and active in management domain
Fa0/1	99

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	99

```
SW1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	NVRAM	up	down
Vlan99	192.168.99.11	YES	NVRAM	up	up
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	administratively down	down
FastEthernet0/4	unassigned	YES	unset	administratively down	down
FastEthernet0/5	unassigned	YES	unset	administratively down	down
FastEthernet0/6	unassigned	YES	unset	administratively down	down
FastEthernet0/7	unassigned	YES	unset	administratively down	down
FastEthernet0/8	unassigned	YES	unset	administratively down	down
FastEthernet0/9	unassigned	YES	unset	administratively down	down
FastEthernet0/10	unassigned	YES	unset	administratively down	down
FastEthernet0/11	unassigned	YES	unset	administratively down	down
FastEthernet0/12	unassigned	YES	unset	administratively down	down
FastEthernet0/13	unassigned	YES	unset	administratively down	down
FastEthernet0/14	unassigned	YES	unset	administratively down	down
FastEthernet0/15	unassigned	YES	unset	administratively down	down
FastEthernet0/16	unassigned	YES	unset	administratively down	down
FastEthernet0/17	unassigned	YES	unset	administratively down	down
FastEthernet0/18	unassigned	YES	unset	administratively down	down
FastEthernet0/19	unassigned	YES	unset	administratively down	down
FastEthernet0/20	unassigned	YES	unset	administratively down	down
FastEthernet0/21	unassigned	YES	unset	administratively down	down
FastEthernet0/22	unassigned	YES	unset	administratively down	down
FastEthernet0/23	unassigned	YES	unset	administratively down	down
FastEthernet0/24	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down

```
SW1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
```

```
SW1#show running-config | section line vty
line vty 0 4
  access-class MGMT_ONLY in
  exec-timeout 5 0
  logging synchronous
  login local
  transport input ssh
line vty 5 15
  access-class MGMT_ONLY in
  exec-timeout 5 0
  logging synchronous
  login local
  transport input ssh
SW1#
```

```
-----
SW1#show access-list MGMT_ONLY
Standard IP access list MGMT_ONLY
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (2 matches)
 20 deny    any log
```


SW2:

```
SW2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/1, Gi0/2
99	MGMT	active	
999	PARKING_LOT	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24

```
SW2#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	99

Port	Vlans allowed and active in management domain
Fa0/1	99

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	99

```
SW2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Vlan1	unassigned	YES	NVRAM	up	down
Vlan99	192.168.99.12	YES	NVRAM	up	up
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	administratively down	down
FastEthernet0/3	unassigned	YES	unset	administratively down	down
FastEthernet0/4	unassigned	YES	unset	administratively down	down
FastEthernet0/5	unassigned	YES	unset	administratively down	down
FastEthernet0/6	unassigned	YES	unset	administratively down	down
FastEthernet0/7	unassigned	YES	unset	administratively down	down
FastEthernet0/8	unassigned	YES	unset	administratively down	down
FastEthernet0/9	unassigned	YES	unset	administratively down	down
FastEthernet0/10	unassigned	YES	unset	administratively down	down
FastEthernet0/11	unassigned	YES	unset	administratively down	down
FastEthernet0/12	unassigned	YES	unset	administratively down	down
FastEthernet0/13	unassigned	YES	unset	administratively down	down
FastEthernet0/14	unassigned	YES	unset	administratively down	down
FastEthernet0/15	unassigned	YES	unset	administratively down	down
FastEthernet0/16	unassigned	YES	unset	administratively down	down
FastEthernet0/17	unassigned	YES	unset	administratively down	down
FastEthernet0/18	unassigned	YES	unset	administratively down	down
FastEthernet0/19	unassigned	YES	unset	administratively down	down
FastEthernet0/20	unassigned	YES	unset	administratively down	down
FastEthernet0/21	unassigned	YES	unset	administratively down	down
FastEthernet0/22	unassigned	YES	unset	administratively down	down
FastEthernet0/23	unassigned	YES	unset	administratively down	down
FastEthernet0/24	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down

```
SW2#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 60 secs; Authentication retries: 3
```

```
SW2#show running-config | section line vty
line vty 0 4
  access-class MGMT_ONLY in
  exec-timeout 5 0
  logging synchronous
  login local
  transport input ssh
line vty 5 15
  access-class MGMT_ONLY in
  exec-timeout 5 0
  logging synchronous
  login local
  transport input ssh
```

```
SW2#show access-lists MGMT_ONLY
Standard IP access list MGMT_ONLY
 10 permit 192.168.99.0, wildcard bits 0.0.0.255 (2 matches)
 20 deny any log
```